Semantic Exploration Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Analysis of Adjacent Semantic Distance

PENPAL Project

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Contents

1 Overview

The semantic exploration outputs contain the **current lag-based structure**:

- `k`: lag
- `distance`: mean semantic distance for that lag
- `agent`: `"interleaved"` for the full exchange sequence, or `author_1` / `author_2` for speaker-specific trajectories

The processed filename is still `semantic_exploration_binned.parquet`, but the current generator computes **lag-based distances**, not the older non-overlapping window-centroid bins.

This notebook uses the raw interaction embeddings to reconstruct the alternating dyadic sequence and analyse:

1. **Adjacent turn-to-turn semantic distance** across conditions as the primary analysis

2. **Story-level mean adjacent distance** as the supporting aggregate
3. **Turn-level adjacent semantic-distance networks** as descriptive 2D maps
4. **Lag-based trajectories** as supplemental descriptive context

Important comparison note:

- The adjacent sequence includes both within-exchange steps (e.g., slot 1_t to slot 2_t) and between-exchange steps (e.g., slot 2_t to slot 1_{t+1}).
- For Human-AI, the sequence is kept in the standard human-to-AI row order. For same-type dyads, `starter_side` determines which slot speaks first within each exchange.
- **Human / AI** in Human-AI and **Starter-side / Non-starter-side** in same-type conditions are only used as descriptive views.

Primary cross-condition object. Each adjacent pair in the reconstructed alternating sequence contributes one semantic distance. The main model compares these adjacent distances by condition while nesting observations within story:

$$d_{ij} = \beta_0 + \beta_1 \text{condition}_{ij} + u_{\text{story}(j)} + \epsilon_{ij}.$$

Article-use note. This analysis answers whether adjacent turns move through semantic space differently by condition. It is not a directional slot-asymmetry test.

Network-map note. The 2D semantic-network layouts are precomputed by `scripts/09_compute_semantic_network_layo` this notebook only reads and visualizes those files. By default, the network script samples the same number of stories and adjacent edges per condition for visualization only, so panels are not visually dominated by sample size.

2 Load Data

Table 1: Loaded semantic exploration rows by condition and agent view

condition	agent	n_stories	n_rows	min_k	max_k
Human-AI	author_1	97	577	2	7
Human-AI	author_2	97	577	2	7
Human-AI	interleaved	97	873	2	10
Human-Human	author_1	36	252	2	8
Human-Human	author_2	36	252	2	8
Human-Human	interleaved	36	324	2	10
AI-AI	author_1	80	560	2	8
AI-AI	author_2	80	560	2	8
AI-AI	interleaved	80	720	2	10

3 Preprocessing

```
## Interleaved rows: 1917
```

```
## Speaker rows: 2778
```

```
## Embedding rows: 1836
```

4 Primary: Adjacent Turn-to-Turn Semantic Distance

Table 2: Adjacent semantic-distance observations by condition

condition	n_adjacent_pairs	n_stories
Human-AI	1453	97
Human-Human	646	36
AI-AI	1360	80

Table 3: Adjacent turn-to-turn semantic distance by condition

condition	n	n_stories	mean_distance	sd_distance	median_distance
Human-AI	1453	97	0.316	0.067	0.313
Human-Human	646	36	0.287	0.065	0.283
AI-AI	1360	80	0.296	0.064	0.292

```
## Main mixed model: adjacent_distance ~ condition + (1 | conversation_id)
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: adjacent_distance ~ condition + (1 | conversation_id)
##    Data: df_adjacent
##
## REML criterion at convergence: -9258.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.1663 -0.6903 -0.0557  0.6429  3.2531
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0006419 0.02534
##  Residual                    0.0036819 0.06068
## Number of obs: 3459, groups:  conversation_id, 213
##
## Fixed effects:
##                        Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)            0.316309   0.003025 217.177645 104.556  < 2e-16 ***
## conditionHuman-Human  -0.029879   0.005718 203.028618  -5.225 4.30e-07 ***
## conditionAI-AI        -0.019818   0.004459 209.411367  -4.444 1.43e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H
## cndtnHmn-Hm -0.529
## condtnAI-AI -0.678  0.359
```

```
##
## Estimated marginal means by condition:
##  condition   emmean      SE  df asymp.LCL asymp.UCL
##  Human-AI     0.316 0.00303 Inf     0.310     0.322
##  Human-Human  0.286 0.00485 Inf     0.277     0.296
##  AI-AI        0.296 0.00328 Inf     0.290     0.303
##
```

```
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
## Pairwise condition contrasts (BH-adjusted):
##  contrast                    estimate      SE  df z.ratio p.value
##  (Human-AI) - (Human-Human)    0.0299 0.00572 Inf   5.225  <.0001
##  (Human-AI) - (AI-AI)          0.0198 0.00446 Inf   4.444  <.0001
##  (Human-Human) - (AI-AI)      -0.0101 0.00585 Inf  -1.719  0.0857
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: BH method for 3 tests
##
## Planned contrast: Human-AI vs same-type average:
##  contrast                             estimate      SE  df z.ratio p.value
##  Human-AI - mean(Human-Human, AI-AI)    0.0248 0.00421 Inf   5.903  <.0001
##
## Degrees-of-freedom method: asymptotic
```

5 Supporting: Story-Level Mean Adjacent Distance

Table 4: Story-level mean adjacent semantic distance by condition

condition	n	mean_story_distance	sd_story_distance	median_story_distance
Human-AI	97	0.316	0.033	0.316
Human-Human	36	0.286	0.023	0.283
AI-AI	80	0.296	0.028	0.297

```
## One-way ANOVA: mean_adjacent_distance ~ condition
##              Df  Sum Sq  Mean Sq F value   Pr(>F)
## condition     2 0.03044 0.015218   17.36 1.06e-07 ***
## Residuals   210 0.18412 0.000877
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Kruskal-Wallis (robust): mean_adjacent_distance ~ condition
##
##  Kruskal-Wallis rank sum test
##
## data:  mean_adjacent_distance by condition
## Kruskal-Wallis chi-squared = 28.507, df = 2, p-value = 6.454e-07
##
##
## Pairwise Wilcoxon (BH-adjusted):
##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  df_adjacent_story$mean_adjacent_distance and df_adjacent_story$condition
##
##             Human-AI Human-Human
## Human-Human 9e-06    -
## AI-AI       0.00014  0.04527
##
```

```
## P value adjustment method: BH
```

6 Bootstrap CIs and Effect Sizes on Story Means

Table 5: Bootstrap 95% CI on per-condition mean of story-level adjacent distance (R = 10,000)

condition	n	mean	ci_lower	ci_upper
Human-AI	97	0.316	0.310	0.323
Human-Human	36	0.286	0.279	0.294
AI-AI	80	0.296	0.290	0.303

Table 6: Pairwise Cliff's delta on story-level mean adjacent distance

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	97	36	0.529	large
Human-AI vs AI-AI	97	80	0.341	medium
Human-Human vs AI-AI	36	80	-0.233	small

7 Planned Contrast: HA vs (HH + AA)/2

Table 7: Planned contrast: story-level mean adjacent distance in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.025	0.004	5.965	168.89	0	0.017	0.033

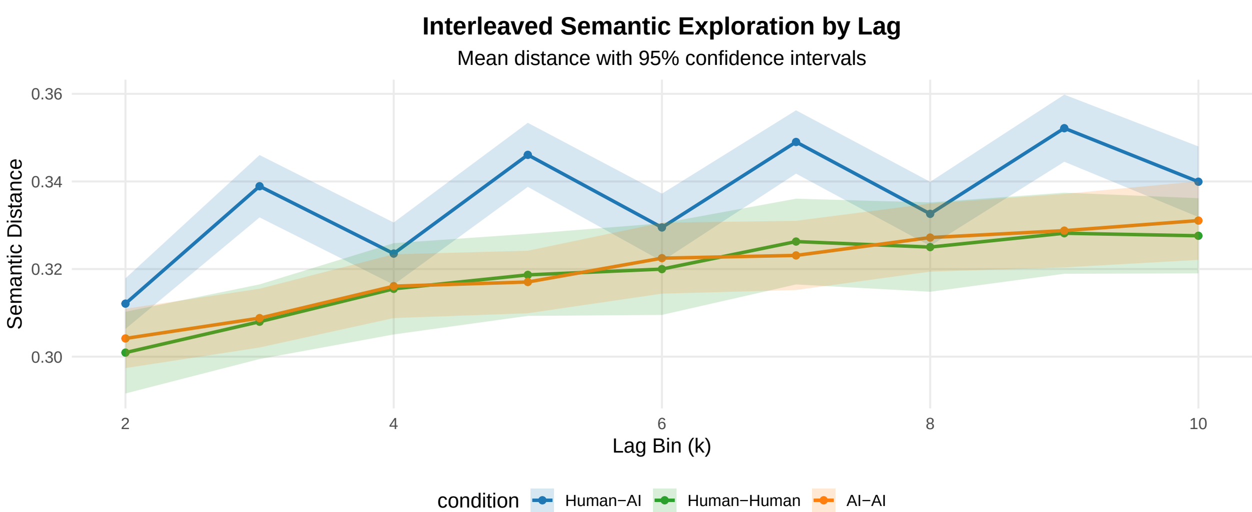
8 Supplemental: Turn-Level Adjacent Semantic-Distance Network

9 Supplemental: General Condition-Level Comparison

Table 8: Interleaved semantic distance by lag and condition

condition	k	mean_distance	sd_distance	n
Human-AI	2	0.312	0.029	97
Human-AI	3	0.339	0.036	97
Human-AI	4	0.324	0.036	97
Human-AI	5	0.346	0.037	97
Human-AI	6	0.329	0.039	97
Human-AI	7	0.349	0.036	97
Human-AI	8	0.333	0.037	97
Human-AI	9	0.352	0.038	97
Human-AI	10	0.340	0.040	97

condition	k	mean_distance	sd_distance	n
Human-Human	2	0.301	0.029	36
Human-Human	3	0.308	0.026	36
Human-Human	4	0.315	0.032	36
Human-Human	5	0.319	0.029	36
Human-Human	6	0.320	0.032	36
Human-Human	7	0.326	0.030	36
Human-Human	8	0.325	0.031	36
Human-Human	9	0.328	0.028	36
Human-Human	10	0.328	0.026	36
AI-AI	2	0.304	0.031	80
AI-AI	3	0.309	0.031	80
AI-AI	4	0.316	0.033	80
AI-AI	5	0.317	0.033	80
AI-AI	6	0.322	0.037	80
AI-AI	7	0.323	0.036	80
AI-AI	8	0.327	0.035	80
AI-AI	9	0.329	0.038	80
AI-AI	10	0.331	0.041	80




```
## Interleaved model summary:

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: distance ~ k * condition + (1 | conversation_id)
##    Data: df_interleaved
##
## REML criterion at convergence: -9389.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.7835 -0.5532  0.0144  0.5725  3.3368
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0009747 0.03122
##  Residual                    0.0002878 0.01697
```

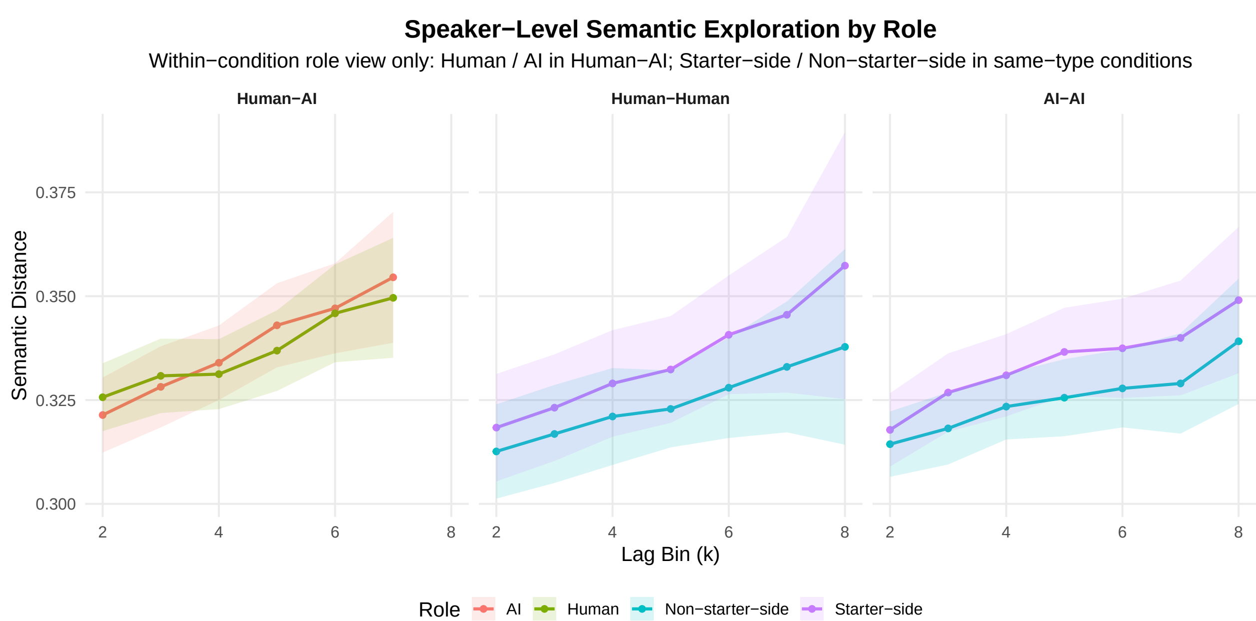
```
## Number of obs: 1917, groups:  conversation_id, 213
##
## Fixed effects:
##                          Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)             3.188e-01  3.487e-03  2.872e+02  91.415  < 2e-16 ***
## k                       2.869e-03  2.224e-04  1.701e+03  12.902  < 2e-16 ***
## conditionHuman-Human   -1.926e-02  6.702e-03  2.872e+02  -2.873 0.004366 **
## conditionAI-AI         -1.848e-02  5.187e-03  2.872e+02  -3.564 0.000428 ***
## k:conditionHuman-Human  3.634e-04  4.275e-04  1.701e+03   0.850 0.395325
## k:conditionAI-AI        3.938e-04  3.308e-04  1.701e+03   1.191 0.233990
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) k      cndH-H cAI-AI k:cH-H
## k           -0.383
## cndtnHmn-Hm -0.520  0.199
## condtnAI-AI -0.672  0.257  0.350
## k:cndtnHm-H  0.199 -0.520 -0.383 -0.134
## k:cndtAI-AI  0.257 -0.672 -0.134 -0.383  0.350

##
## Condition contrasts (interleaved view):

##  contrast                     estimate      SE  df t.ratio p.value
##  (Human-AI) - (Human-Human)  0.017077 0.00619 210   2.758  0.0190
##  (Human-AI) - (AI-AI)        0.016122 0.00479 210   3.364  0.0027
##  (Human-Human) - (AI-AI)    -0.000955 0.00637 210  -0.150  1.0000
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

10 Supplemental: Role-Level View

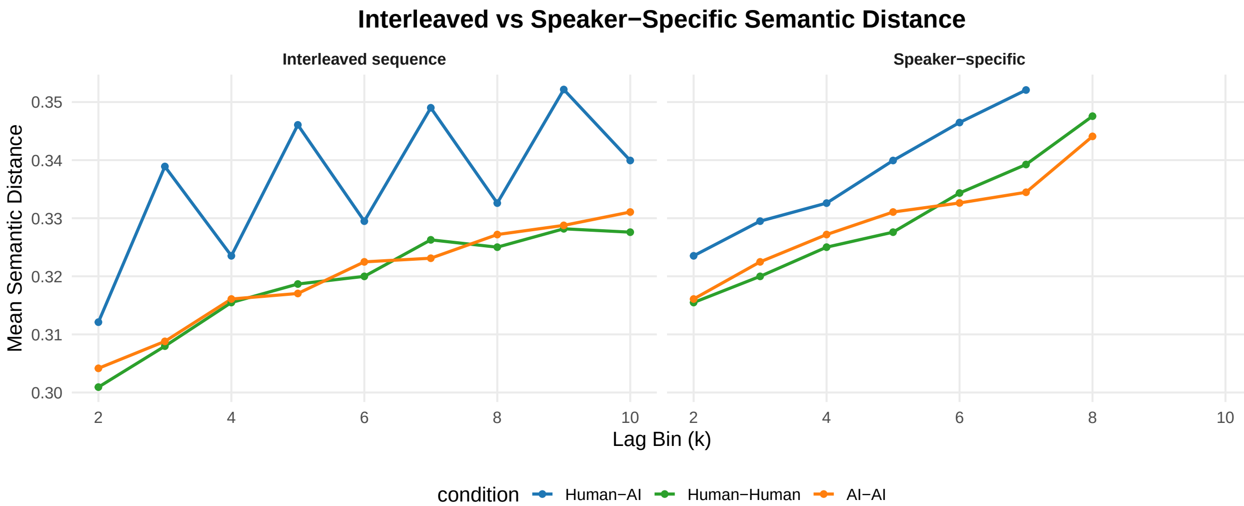
Table 9: Speaker-level semantic distance by condition, role, and lag

condition	condition_role	k	mean_distance	n
Human-AI	AI	2	0.321	97
Human-AI	AI	3	0.328	97
Human-AI	AI	4	0.334	97
Human-AI	AI	5	0.343	97
Human-AI	AI	6	0.347	96
Human-AI	AI	7	0.355	93
Human-AI	Human	2	0.326	97
Human-AI	Human	3	0.331	97
Human-AI	Human	4	0.331	97
Human-AI	Human	5	0.337	97
Human-AI	Human	6	0.346	96
Human-AI	Human	7	0.350	93
Human-Human	Non-starter-side	2	0.313	36
Human-Human	Non-starter-side	3	0.317	36
Human-Human	Non-starter-side	4	0.321	36

condition	condition_role	k	mean_distance	n
Human-Human	Non-starter-side	5	0.323	36
Human-Human	Non-starter-side	6	0.328	36
Human-Human	Non-starter-side	7	0.333	36
Human-Human	Non-starter-side	8	0.338	36
Human-Human	Starter-side	2	0.318	36
Human-Human	Starter-side	3	0.323	36
Human-Human	Starter-side	4	0.329	36
Human-Human	Starter-side	5	0.332	36
Human-Human	Starter-side	6	0.341	36
Human-Human	Starter-side	7	0.346	36
Human-Human	Starter-side	8	0.357	36
AI-AI	Non-starter-side	2	0.314	80
AI-AI	Non-starter-side	3	0.318	80
AI-AI	Non-starter-side	4	0.323	80
AI-AI	Non-starter-side	5	0.326	80
AI-AI	Non-starter-side	6	0.328	80
AI-AI	Non-starter-side	7	0.329	80
AI-AI	Non-starter-side	8	0.339	80
AI-AI	Starter-side	2	0.318	80
AI-AI	Starter-side	3	0.327	80
AI-AI	Starter-side	4	0.331	80
AI-AI	Starter-side	5	0.337	80
AI-AI	Starter-side	6	0.337	80
AI-AI	Starter-side	7	0.340	80
AI-AI	Starter-side	8	0.349	80



11 Supplemental: Interleaved vs Speaker View



12 Summary

Primary cross-condition result: adjacent turn-to-turn semantic distance from the reconstructed alternating dyadic sequence, tested with a mixed model that nests adjacent observations within story. This includes both within-exchange distances and the cross-exchange step from the second turn of one exchange to the first turn of the next exchange.

Robustness layers added:

- story-level mean adjacent distance as an aggregate supporting analysis
- bootstrap 95% CIs on per-condition story means
- pairwise Cliff's δ effect sizes on story means
- planned single-df contrast HA vs (HH+AA)/2 on story means

Supplemental views retained:

- turn-level adjacent semantic-distance network, with edge color/width encoding original-space adjacent distance
- lag-based speaker-level slot trajectories
- lag-based interleaved sequence model and condition descriptives
- within-condition role curves (descriptive only; HA Human/AI inferential view lives in analysis/human-ai/exploration.Rmd)
- interleaved vs speaker-specific side-by-side view

To refresh the semantic-network layouts, run `venv/bin/python scripts/09_compute_semantic_network_layouts.py` before rendering this notebook.

Analysis completed: 2026-05-18 15:45:52.154822